SOK-3081 | Chapter 5 seminar

Questions adapted from Hill, Griffiths and Lim, *Principles of Econometrics*, 5th edition. Use the matching worked-solutions document after attempting them.

Exercise 5.1: Estimate a multiple regression by hand

Consider $y_i = x_{i1}\beta_1 + x_{i2}\beta_2 + x_{i3}\beta_3 + e_i$, with the following seven observations (textbook Table 5.5).

i	y_i	x_{i1}	x_{i2}	x_{i3}
1	1	1	0	1
2	1	1	1	-2
3	4	1	2	2
4	0	1	-2	1
5	1	1	1	-2
6	-2	1	-2	-1
7	2	1	0	1

Use a calculator or spreadsheet.

(a) Express the observations as deviations from their sample means: $x^*_{i2} = x_{i2} - \bar{x}_2$, $x^*_{i3} = x_{i3} - \bar{x}_3$, and $y^*_i = y_i - \bar{y}$.

(b) Calculate $\sum_i y^*_i x^*_{i2}$, $\sum_i (x^*_{i2})^2$, $\sum_i y^*_i x^*_{i3}$, $\sum_i x^*_{i2} x^*_{i3}$, and $\sum_i (x^*_{i3})^2$.

(c) Use the formulas in Appendix 5A to find b_1, b_2, and b_3.

(d) Find the seven least-squares residuals.

(e) Estimate the error variance σ^2.

(f) Find the sample correlation between x_2 and x_3.

(g) Find the standard error of b_2.

(h) Find SSE, SST, SSR, and R^2.

Hints for Exercise 5.1

(b) These are the familiar cross-products and sums of squares. Use the short notation

$$\begin{aligned}
\sum_i y_i^* x_{i2}^* &= S_{y2}, \quad \sum_i (x_{i2}^*)^2 = S_{22}, \\
\sum_i y_i^* x_{i3}^* &= S_{y3}, \quad \sum_i x_{i2}^* x_{i3}^* = S_{23}, \\
\sum_i (x_{i3}^*)^2 &= S_{33}.
\end{aligned}$$

First calculate each product for every observation; then sum down each column.

(c) Define the common denominator

$$D = S_{22}S_{33} - S_{23}^2.$$

Then use

$$\begin{aligned}
b_2 &= \frac{S_{y2}S_{33} - S_{y3}S_{23}}{D}, \\
b_3 &= \frac{S_{y3}S_{22} - S_{y2}S_{23}}{D}, \\
b_1 &= \bar{y} - b_2\bar{x}_2 - b_3\bar{x}_3.
\end{aligned}$$

(d) For each observation, first calculate the fitted value and then the residual:

$$\hat{y}_i = b_1 + b_2 x_{i2} + b_3 x_{i3}, \qquad \hat{e}_i = y_i - \hat{y}_i.$$

(e) Square and add the residuals from part (d), then divide by the residual degrees of freedom:

$$SSE = \sum_{i=1}^{N} \hat{e}_i^2, \qquad \hat{\sigma}^2 = \frac{SSE}{N - K}.$$

Here N is the number of observations and K is the number of estimated coefficients, including the intercept.

(f) Use the quantities calculated in part (b):

$$r_{23} = \frac{S_{23}}{\sqrt{S_{22}S_{33}}}.$$

(g) First find the estimated variance of b_2, then take its square root:

$$\widehat{\mathrm{var}}(b_2 \mid \mathbf{X}) = \frac{\hat{\sigma}^2}{(1 - r_{23}^2)S_{22}}, \qquad se(b_2) = \sqrt{\widehat{\mathrm{var}}(b_2 \mid \mathbf{X})}.$$

(h) Use

$$
\begin{aligned}
SSE &= \sum_i \hat{e}_i^2, \\
SST &= \sum_i (y_i - \bar{y})^2, \\
SSR &= SST - SSE, \\
R^2 &= \frac{SSR}{SST} = 1 - \frac{SSE}{SST}.
\end{aligned}
$$

Exercise 5.2: Confidence interval and hypothesis test

Use your results from Exercise 5.1.

(a) Compute a 95% confidence interval for β_2.

(b) Test $H_0 : \beta_2 = 1.25$ against $H_1 : \beta_2 \neq 1.25$. State the significance level you use and your conclusion.

Hints for Exercise 5.2

(a) Use the estimate and standard error from Exercise 5.1:

$$b_2 \pm t_{0.975,\,N-K}\, se(b_2).$$

There are $N - K$ residual degrees of freedom. Find the corresponding critical value from the t distribution before calculating the lower and upper endpoints.

(b) This is a two-sided test. The test statistic is

$$t = \frac{b_2 - 1.25}{se(b_2)}.$$

At the 5% significance level, compare $|t|$ with $t_{0.975,\,N-K}$. Reject H_0 if $|t|$ is larger than the critical value. Equivalently, calculate the two-sided p-value $2P(T_{N-K} > |t|)$ and compare it with 0.05. As a final check, ask whether 1.25 lies inside the 95% confidence interval from part (a).

Exercise 5.3: Complete and interpret regression output

For 1200 London households, estimate $WALC = \beta_1 + \beta_2 \ln(TOTEXP) + \beta_3 NK + \beta_4 AGE + e$. $WALC$ is the percentage of the budget spent on alcohol; $TOTEXP$ is total expenditure, NK the number of children, and AGE the household head's age. The supplied output (Table 5.6) is incomplete.

Variable	Coefficient	Standard error	t statistic	p-value
Intercept	1.4515	2.2019	?	0.5099
$\ln(TOTEXP)$	2.7648	?	5.7103	0.0000
NK	?	0.3695	-3.9376	0.0001
AGE	-0.1503	0.0235	-6.4019	0.0000

Additional output: $SSE = 46221.62$, sample mean of $WALC = 6.19434$, and sample standard deviation of $WALC = 6.39547$.

(a) Find the missing intercept t statistic, standard error of b_2, estimate b_3, R^2, and regression standard error $\hat{\sigma}$.

(b) Interpret b_2, b_3, and b_4.

(c) Compute a 95% confidence interval for β_4. Interpret it.

(d) Which coefficients differ significantly from zero at 5%? Explain.

(e) At 5%, test whether an additional child reduces the mean alcohol budget share by 2 percentage points against a two-sided alternative.

Hints for Exercise 5.3

There are $N = 1200$ observations and $K = 4$ estimated coefficients, including the intercept. Thus, use $N - K$ residual degrees of freedom.

(a) Complete the regression output by rearranging the usual t-statistic:

$$t_k = \frac{b_k}{se(b_k)}, \qquad se(b_k) = \frac{b_k}{t_k}, \qquad b_k = t_k\, se(b_k).$$

The reported sample standard deviation uses $N - 1$ in its denominator. For R^2 and the regression standard error, use

$$SST = (N-1)s_y^2, \qquad R^2 = 1 - \frac{SSE}{SST}, \qquad \hat{\sigma} = \sqrt{\frac{SSE}{N-K}}.$$

(b) Interpret every coefficient while holding the other included variables constant. Since $\ln(TOTEXP)$ is used, a 1% increase in $TOTEXP$ changes mean $WALC$ by approximately $b_2/100$ **percentage points**. For NK and AGE, interpret a one-unit increase directly. Pay attention to the signs and distinguish percentage points from percent.

(c) Use

$$b_4 \pm t_{0.975,\, N-K}\, se(b_4).$$

Interpret both endpoints as changes in the mean alcohol budget share, in percentage points per additional year of age.

(d) Each reported p-value tests $H_0 : \beta_k = 0$ against $H_1 : \beta_k \neq 0$. At the 5% level, reject H_0 when the p-value is at most 0.05. Equivalently, compare $|t_k|$ with the two-sided critical value.

(e) "Reduces by 2 percentage points" corresponds to $H_0 : \beta_3 = -2$. Use the two-sided statistic

$$t = \frac{b_3 - (-2)}{se(b_3)} = \frac{b_3 + 2}{se(b_3)}.$$

Compare $|t|$ with $t_{0.975,\, N-K}$, or compare the two-sided p-value with 0.05. Remember that testing $\beta_3 = -2$ is not the same as testing $\beta_3 = 0$.

Exercise 5.4: Alcohol expenditure, children and age

For 1200 households, the model is $WALC = \beta_1 + \beta_2 \ln(TOTEXP) + \beta_3 NK + \beta_4 AGE + \beta_5 AGE^2 + e$. Definitions are as in Exercise 5.3. Table 5.7 supplies $\widehat{WALC} = 8.149 + 2.884 \ln(TOTEXP) - 1.217NK - 0.5699AGE + 0.005515AGE^2$.

The relevant estimated covariance matrix is:

	b_3 (NK)	b_4 (AGE)	b_5 (AGE^2)
b_3	0.1462	-0.01774	0.0002347
b_4	-0.01774	0.03204	-0.0004138
b_5	0.0002347	-0.0004138	0.000005438

(a) Find a point estimate and 95% confidence interval for the change in mean alcohol budget share from one additional child.

(b) Find point estimates and 95% confidence intervals for the marginal effect of age at ages (i) 25, (ii) 50, and (iii) 75.

(c) Find a point estimate and 95% confidence interval for the age at which the mean alcohol budget share is minimized.

(d) Summarize what (a)-(c) tell you.

(e) If errors conditional on $\mathbf{X}$ are normal, which intervals are valid in finite samples, and which require a large-sample approximation?

(f) Answer (e) if the conditional errors are not normal.

Hints for Exercise 5.4

There are $N = 1200$ observations and $K = 5$ estimated coefficients, including the intercept. Use $N - K$ residual degrees of freedom for the t critical value.

(a) The effect of one additional child is β_3. Read its point estimate from the fitted equation. Its variance is the diagonal covariance matrix entry for b_3:

$$se(b_3) = \sqrt{\widehat{\text{var}}(b_3)}, \qquad b_3 \pm t_{0.975,\, N-K} se(b_3).$$

(b) Because both AGE and AGE^2 are included, the marginal effect at age a is

$$m(a) = \frac{\partial E(WALC \mid \mathbf{X})}{\partial AGE} = \beta_4 + 2a\beta_5.$$

Estimate it using $\widehat{m}(a) = b_4 + 2ab_5$. For its standard error, use the variance formula for a linear combination:

$$\widehat{\text{var}}[\widehat{m}(a)] = \widehat{\text{var}}(b_4) + 4a^2\widehat{\text{var}}(b_5) + 4a\widehat{\text{cov}}(b_4, b_5).$$

Then calculate

$$\widehat{m}(a) \pm t_{0.975,\, N-K} \sqrt{\widehat{\text{var}}[\widehat{m}(a)]}$$

separately for $a = 25$, 50, and 75.

(c) At the minimum, the marginal effect of age equals zero. Solve $\beta_4 + 2a^*\beta_5 = 0$:

$$\hat{a}^* = -\frac{b_4}{2b_5}.$$

Check that $b_5 > 0$, so the turning point is a minimum. For an approximate standard error, let $g(b_4, b_5) = -b_4/(2b_5)$ and use the delta method with

$$g_4 = -\frac{1}{2b_5}, \qquad g_5 = \frac{b_4}{2b_5^2},$$

and

$$\widehat{\text{var}}(\hat{a}^*) \approx g_4^2\widehat{\text{var}}(b_4) + g_5^2\widehat{\text{var}}(b_5) + 2g_4g_5\widehat{\text{cov}}(b_4, b_5).$$

Take the square root and form the approximate 95% interval in the usual way.

(d) Summarize direction, magnitude, and uncertainty. Ask whether each interval contains zero. Describe the fitted age pattern before and after the turning point, but avoid making a causal claim without further assumptions.

(e) Under conditionally normal errors and MR1–MR6, intervals for linear combinations of coefficients have exact finite-sample t justification. The turning point is a nonlinear ratio of coefficients, so its delta-method interval remains a large-sample approximation.

(f) Without conditional normality, the usual finite-sample t results are not exact. Under the other assumptions and suitable large-sample conditions, all three interval procedures can instead be justified approximately.

Exercise 5.5: Does additional data ensure consistency?

Assume MR1-MR6. For each model, find $\text{var}(b_2 \mid \mathbf{x})$ and examine consistency by checking whether this variance tends to zero as $T \to \infty$.

(a) $y_t = \beta_1 + \beta_2 t + e_t$, for $t = 1, \ldots, T$.

Useful identities: $\sum_{t=1}^{T} t = T(T+1)/2$, $\sum_{t=1}^{T} t^2 = T(T+1)(2T+1)/6$, and $\sum_t (t - \bar{t})^2 = \sum_t t^2 - (\sum_t t)^2/T$.

(b) $y_t = \beta_1 + \beta_2 (0.5)^t + e_t$, for $t = 1, \ldots, T$.

Useful identity: $r + r^2 + \cdots + r^n = r(1 - r^n)/(1 - r)$.

(c) Give an intuitive explanation for your results.

Exercise 5.6: Test linear combinations

For a regression with three coefficients and $N = 63$, the estimates are $b_1 = 2$, $b_2 = 3$, and $b_3 = -1$. Their estimated covariance matrix is:

	b_1	b_2	b_3
b_1	3	-2	1
b_2	-2	4	0
b_3	1	0	3

Use a 5% significance level and a two-sided alternative to test:

(a) $H_0 : \beta_2 = 0$.

(b) $H_0 : \beta_1 + 2\beta_2 = 5$.

(c) $H_0 : \beta_1 - \beta_2 + \beta_3 = 4$.

Exercise 5.7: Precision, correlation and tests

For $y = \beta_1 + \beta_2 x_2 + \beta_3 x_3 + e$, $N = 203$. You are given $\sum_i (x_{i2} - \bar{x}_2)^2 = 1780.7$, $\sum_i (x_{i3} - \bar{x}_3)^2 = 3453.3$, $b_2 = 0.7176$, $b_3 = 1.0516$, $SSE = 6800.0$, and $r_{23} = 0.7087$.

(a) Find the standard errors of b_2 and b_3.

(b) At 5%, test $H_0 : \beta_2 = 0$ against $H_1 : \beta_2 \neq 0$.

(c) At 10%, test $H_0 : \beta_3 \leq 0.9$ against $H_1 : \beta_3 > 0.9$.

(d) Given $\widehat{\text{cov}}(b_2, b_3) = -0.019521$, test $H_0 : \beta_2 = \beta_3$ against $H_1 : \beta_2 \neq \beta_3$ at 1%.

Exercise 5.8: Medals at the 1996 Olympics

For the 79 countries winning at least one medal, $MEDALS$ is the medal count, $POPM$ population in millions, and $GDPB$ GDP in billions of 1995 dollars. Assume MR1-MR6. The fitted model has $R^2 = 0.4879$: $\widehat{MEDALS} = 5.917 + 0.01813POPM + 0.01026GDPB$. Standard errors, in coefficient order: (1.510, 0.00819, 0.00136).

(a) Interpret the population and GDP coefficient estimates.

(b) Interpret R^2.

(c) Test no relationship with GDP against a positive relationship at 1%. Does the conclusion change at 5%?

(d) Repeat (c) for population.

(e) At 5%, test $H_0 : \beta_2 = c$ against $H_1 : \beta_2 \neq c$ for (i) $c = 0.01$, (ii) 0.02, (iii) 0.03, and (iv) 0.04. Are the results contradictory? Explain.

(f) Find a 95% confidence interval for β_2 and comment on it.

Exercise 5.9: Medals at the 1992 Olympics

Among the 64 medal-winning countries, exclude the United Kingdom and estimate with $N = 63$. Definitions are as in Exercise 5.8. Assume MR1-MR6. The fitted model has $R^2 = 0.275$: $\widehat{MEDALS} = -13.153 + 2.764\ln(POPM) + 4.270\ln(GDPB)$. Standard errors: (5.974, 2.070, 1.718).

(a) Interpret the population and GDP coefficients.

(b) Interpret R^2.

(c) Test no relationship with GDP against a positive relationship at 10%. What happens at 5%?

(d) Repeat (c) for population.

(e) Find point and 95% interval estimates for the UK's expected medal count, with $POPM = 58$ and $GDPB = 1010$. The standard error of $b_1 + \ln(58)b_2 + \ln(1010)b_3$ is 4.22196.

(f) The UK won 20 medals. Is the model useful for predicting its mean medal count? Find an approximate two-sided p-value for the null that its expected count equals 20.

(g) Write the expression for the standard error in (e), without evaluating it.

Exercise 5.10: Wheat yield and rainfall

For Mullewa, Western Australia, $T = 47$ annual observations (1950-1996) give $\widehat{YIELD}_t = 0.1717 + 0.01117t + 0.05238RAIN_t$. Standard errors: (0.1537, 0.00262, 0.01367). Yield is tonnes per hectare, $t = 1, \ldots, 47$, and rainfall is growing-season rainfall in inches. Its sample mean is 10.059 and standard deviation 2.624.

(a) Assuming MR1-MR5, interpret the trend and rainfall coefficients.

(b) At 5%, test whether technological change increases mean annual yield by no more than 0.01 tonnes per hectare against a larger increase.

(c) At 5%, test whether an extra inch of rain increases mean yield by 0.03 tonnes per hectare against a two-sided alternative.

(d) Adding squared rainfall gives $\widehat{YIELD}_t = -0.6759 + 0.011671t + 0.2229RAIN_t - 0.008155RAIN_t^2$. Standard errors: (0.3875, 0.00250, 0.0734, 0.003453). Why include squared rainfall? Does its coefficient have the expected sign?

(e) Repeat (b) using the quadratic model.

(f) Repeat (c) at mean rainfall using the quadratic model. For its rainfall coefficients, $\widehat{\text{cov}}(b_3, b_4) = -0.0002493$.

(g) Forecast yield in 1997 when rainfall was 9.48 inches, using (d).

(h) Before observing 1997 rainfall, what forecast would you make using each model? State your assumptions about rainfall.

Exercise 5.11: A life-cycle wage profile

Consider $WAGE = \beta_1 + \beta_2 EDUC + \beta_3 EXPER + \beta_4 EXPER^2 + e$. Education and experience are measured in years.

(a) Find the marginal effect of experience on mean wage.

(b) What signs do you expect for β_2, β_3, and β_4? Explain.

(c) Express in terms of the parameters the experience level after which mean wage begins to decline.

(d) Using 600 observations, the fitted equation is $\widehat{WAGE} = -16.308 + 2.329 EDUC + 0.5240 EXPER - 0.007582 EXPER^2$. Standard errors: (2.745, 0.163, 0.1263, 0.002532). The estimated covariance of the experience coefficients is $\widehat{\text{cov}}(b_3, b_4) = -0.00030526$. Find 95% confidence intervals for:

- **(i)** The marginal effect of education on mean wage.
- **(ii)** The marginal effect of experience at $EXPER = 4$.
- **(iii)** The marginal effect of experience at $EXPER = 25$.
- **(iv)** The experience level after which mean wage declines.

Exercise 5.12: House prices and residual regressions

For 850 houses sold in Baton Rouge in mid-2005, $PRICE$ is in thousands of dollars, $SQFT$ in hundreds of square feet, and AGE in years. Regress $PRICE = \alpha_1 + \alpha_2 AGE + v$ and $SQFT = \delta_1 + \delta_2 AGE + u$. The fitted residuals satisfy $\sum_i \hat{v}_i^2 = 10377817$, $\sum_i \hat{u}_i^2 = 75773.4$, and $\sum_i \hat{u}_i \hat{v}_i = 688318$.

(a) Find the OLS estimate of β_2 in $PRICE = \beta_1 + \beta_2 SQFT + \beta_3 AGE + e$.

(b) Let $\hat{e}_i = \hat{v}_i - b_2 \hat{u}_i$. Show that $\sum_i \hat{e}_i^2 = \sum_i \hat{v}_i^2 - b_2 \sum_i \hat{v}_i \hat{u}_i$.

(c) Estimate the error variance σ^2.

(d) Find the standard error of b_2.

(e) Find an approximate p-value for $H_0 : \beta_2 \geq 9.5$ against $H_1 : \beta_2 < 9.5$. What do you conclude?

Exercise 5.13: Okun's law

Okun's law is $U_t - U_{t-1} = -\gamma(G_t - G_N)$, where U_t is the unemployment rate, G_t economic growth, G_N the growth rate needed to keep unemployment constant, and $0 < \gamma < 1$.

(a) Write the model as $DU_t = \beta_1 + \beta_2 G_t$, defining $DU_t = U_t - U_{t-1}$, $\beta_1 = \gamma G_N$, and $\beta_2 = -\gamma$.

(b) Quarterly US data, 1970 Q1-2014 Q4, give $\widehat{DU}_t = 0.1989 - 0.2713 G_t$, with residual standard deviation $\hat{\sigma} = 0.2749$. Find estimates of γ and G_N. The estimated coefficient covariance matrix is:

	b_1	b_2
b_1	0.0007212	-0.0004277
b_2	-0.0004277	0.0006113

(c) Find standard errors of b_1, b_2, $\hat{\gamma}$, and $\widehat{G}_N$. Does each differ significantly from zero at 5%?

(d) At 5%, test whether $G_N = 0.8\%$ per quarter against a two-sided alternative.

(e) Find a 95% confidence interval for the adjustment coefficient.

(f) Find a 95% confidence interval for $E(U_{2015Q1} \mid U_{2014Q4} = 5.7991,\ G_{2015Q1} = 0.062)$.

(g) Find the corresponding 95% prediction interval for U_{2015Q1}. Explain how it differs from (f).

Exercise 5.14: Normality of regressors and errors

Let $y_i = \beta_1 + \beta_2 x_i + e_i$, with independently sampled pairs (y_i, x_i), $i = 1, \ldots, N$.

(a) Suppose x_i is log-normal and $(e_i \mid x_i) \sim N(0, \sigma_e^2)$. If $W \sim N(\mu_W, \sigma_W^2)$, then $X = \exp(W)$ has $\mu_X = \exp(\mu_W + \sigma_W^2/2)$ and $\sigma_X^2 = [\exp(\sigma_W^2) - 1]\mu_X^2$.

- **(i)** Are the OLS estimators of β_1 and β_2 unbiased?
- **(ii)** Are they consistent?
- **(iii)** Are they normally distributed conditional on $\mathbf{x}$?
- **(iv)** Are their unconditional distributions normal?
- **(v)** Are t tests exact in finite samples or only approximate?
- **(vi)** If $\mu_W = 0$, $\sigma_W^2 = 1$, and $x_i = \exp(w_i)$, find the asymptotic variance of b_2 in terms of σ_e^2 and N.

(b) Now suppose $x_i \sim N(0, 1)$ and the conditional error distribution is log-normal, with parameters μ_v, σ_v^2 that do not depend on x_i. Its mean and variance are $\mu_e = \exp(\mu_v + \sigma_v^2/2)$ and $\sigma_e^2 = [\exp(\sigma_v^2) - 1]\mu_e^2$.

- **(i)** Rewrite the model as $y_i = \beta_1^* + \beta_2 x_i + e_i^*$, where $\beta_1^* = \beta_1 + \mu_e$ and $e_i^* = e_i - \mu_e$.
- **(ii)** Show that $E(e_i^* \mid x_i) = 0$ and $\text{var}(e_i^* \mid x_i) = \sigma_e^2$.
- **(iii)** Is b_2 unbiased for β_2?
- **(iv)** Is it consistent?
- **(v)** Is its distribution normal conditional on $\mathbf{x}$?
- **(vi)** Is its unconditional distribution normal?
- **(vii)** Are t tests exact in finite samples or only approximate?
- **(viii)** Find the asymptotic variance of b_2 in terms of σ_e^2 and N.

Exercise 5.15: Zero covariance and exogeneity

Let $y_i = \beta_1 + \beta_2 x_i + e_i$, with independently sampled pairs, $x_i \sim N(0,1)$, and $E(e_i \mid x_i) = c(x_i^2 - 1)$ for a constant c.

(a) Show that $E(e_i) = 0$.

(b) Using $\text{cov}(e_i, x_i) = E_x[(x_i - \mu_x)E(e_i \mid x_i)]$, show that $\text{cov}(e_i, x_i) = 0$.

(c) Is the OLS estimator of β_2 (i) unbiased and (ii) consistent? Explain.

Exercise 5.16: Prices of competing tuna brands

Weekly sales of brand A, $CANS$, are in thousands of cans. Define $RPRCE_B = 100PRICE_B/PRICE_A$ and $RPRCE_C = 100PRICE_C/PRICE_A$. The model is $\ln(CANS) = \beta_1 + \beta_2 RPRCE_B + \beta_3 RPRCE_C + e$.

(a) Assuming MR1-MR5, interpret β_2 and β_3 and discuss their expected signs.

With $N = 52$, the fitted equation is $\widehat{\ln(CANS)} = -2.724 + 0.0146 RPRCE_B + 0.02649 RPRCE_C$. Standard errors: $(0.582,\ 0.00548,\ 0.00544)$; $\hat{\sigma} = 0.5663$ and $\widehat{\text{cov}}(b_2, b_3) = -0.0000143$.

(b) At 10%, test whether a one-percentage-point increase in $RPRCE_B$ increases mean sales by 2.5%, against a two-sided alternative.

(c) Repeat (b) for $RPRCE_C$.

(d) At 10%, test $H_0 : \beta_2 = \beta_3$ against a two-sided alternative. Does this contradict (b)-(c)?

(e) Which competing brand seems the closer substitute for A? Why?

(f) Using the corrected predictor from Section 4.5.3, predict mean brand A sales at $(RPRCE_B, RPRCE_C)$ equal to (i) $(125, 100)$, (ii) $(111.11, 88.89)$, and (iii) $(100, 80)$.

(g) B costs 1 dollar and C costs 80 cents. Which price for A maximizes predicted revenue: 80 cents, 90 cents, or 1 dollar? Use (f).

Exercise 5.17: Check the hand calculations in R

Use R to verify Exercise 5.1, parts **(c), (e), (f), (g), and (h)**. Use the seven observations on the Exercise 5.1 page.

Compare the software results with your hand calculations for:

- The three coefficient estimates.
- The estimated error variance.
- The correlation between the regressors.
- The standard error of b_2.
- SSE, SST, SSR, and R^2.

Exercise 5.18: Food, clothing and household size

Use `london5`. Budget shares satisfy $WFOOD = \beta_1 + \beta_2 \ln(TOTEXP) + e_F$ and $WCLOTH = \alpha_1 + \alpha_2 \ln(TOTEXP) + e_C$. $TOTEXP$ is total expenditure and NK the number of children.

(a) A positive log-expenditure coefficient indicates a luxury; a negative coefficient indicates a necessity. What signs do you expect?

(b) Estimate both equations for one-child households. Discuss the estimates, significance, and any counterintuitive findings.

(c) At 1%, test $H_0 : \beta_2 \geq 0$ against $H_1 : \beta_2 < 0$. Why use this direction?

(d) At 1%, test $H_0 : \alpha_2 \geq 0$ against $H_1 : \alpha_2 < 0$. Why use this direction?

(e) Estimate both equations for two-child households. Compare 95% confidence intervals for the log-expenditure slopes across the one- and two-child groups. Do you expect equal slopes?

(f) Using all observations, estimate $WFOOD = \gamma_1 + \gamma_2 \ln(TOTEXP) + \gamma_3 NK + \gamma_4 NK \ln(TOTEXP) + e_F$ and $WCLOTH = \delta_1 + \delta_2 \ln(TOTEXP) + \delta_3 NK + \delta_4 NK \ln(TOTEXP) + e_C$. Test your conjectures from (e) at 5% (95% confidence).

(g) Compare the marginal log-expenditure effects for food at $NK = 1, 2$ from the separate regressions and the interaction model.

Hints for Exercise 5.18

The student R worksheet downloads `london5` in `START UP`. Use `names(london5)` and `head(london5)` to inspect the variables before estimating the models.

(a) Think about how each budget **share** changes when total expenditure rises. A necessity normally takes a smaller share of the budget as households become richer, whereas a luxury may take a larger share. Translate these ideas into expected signs for the coefficients on $\ln(TOTEXP)$.

(b) First select the one-child households, then estimate the two equations:

```
one_child <- london5 |>
  filter(nk == 1)

food_one <- lm(wfood ~ log(totexp), data = one_child)
clothing_one <- lm(wcloth ~ log(totexp), data = one_child)

summary(food_one)
summary(clothing_one)
```

Use the signs, sizes, standard errors, and p-values when discussing the results. Since the dependent variables are shares, distinguish percent from percentage points.

(c) This is a lower-tail test. Obtain the slope's t-statistic and residual degrees of freedom from the fitted food model. The one-sided p-value is

$$P(T_{N-K} \leq t) = F_{t_{N-K}}(t).$$

In R, `pt(t_value, df)` gives this lower-tail probability. Reject at 1% when the one-sided p-value is below 0.01. Check the sign of the estimate before drawing the conclusion.

(d) Repeat the procedure from (c) for the clothing model. Do not automatically divide a two-sided p-value by two without first checking whether the estimated coefficient points in the direction specified by H_1.

(e) Create a two-child subsample and estimate the same two regressions. Use `confint(model, "log(totexp)", level = 0.95)` for each slope. Comparing separate confidence intervals is descriptive; overlapping intervals do not by themselves constitute a formal test that the two slopes are equal.

(f) With all observations, `*` includes both main effects and the interaction:

```
food_interaction <- lm(wfood ~ log(totexp) * nk, data = london5)
clothing_interaction <- lm(wcloth ~ log(totexp) * nk, data = london5)
```

In each model, test whether the coefficient on `log(totexp):nk` is zero. A zero interaction coefficient means that the log-expenditure slope does not change with the number of children.

(g) In the food interaction model, the marginal effect of $\ln(TOTEXP)$ for a household with $NK = n$ children is

$$\frac{\partial E(WFOOD \mid \mathbf{X})}{\partial \ln(TOTEXP)} = \gamma_2 + n\gamma_4.$$

Evaluate this expression at $n = 1$ and $n = 2$, and compare the resulting slopes with those from the two separate food regressions.

Exercise 5.19: Food expenditure elasticity

Use all observations in `london5` and the model $WFOOD = \beta_1 + \beta_2 \ln(TOTEXP) + e_F$. The elasticity of food expenditure with respect to total expenditure is $\varepsilon = 1 + \beta_2/[\beta_1 + \beta_2 \ln(TOTEXP)]$.

(a) Show that $\beta_2 < 0$ if and only if $\varepsilon < 1$ when the mean food share is positive. Relate this to a necessity.

(b) Estimate the equation and comment on the results.

(c) Find point estimates and 95% confidence intervals for the mean food budget share at $TOTEXP$ equal to (i) 50 (5th percentile), (ii) 90 (median), and (iii) 170 (95th percentile).

(d) Find point estimates and 95% confidence intervals for ε at the same three expenditure levels.

(e) How do the mean share, elasticity and reliability of estimation change with total expenditure?

Exercise 5.20: An alternative slope estimator

For $y_i = \beta_1 + \beta_2 x_i + e_i$, assume MR1-MR6 and even N. Sort observations by increasing x. Let $(\bar{y}_1, \bar{x}_1)$ and $(\bar{y}_2, \bar{x}_2)$ be means for the lower and upper halves. Consider $\widehat{\beta}_{2,mean} = (\bar{y}_2 - \bar{y}_1)/(\bar{x}_2 - \bar{x}_1)$.

(a) Show that this estimator is unbiased.

(b) Derive its conditional variance given $\mathbf{x}$.

(c) Write an expression for its unconditional variance.

(d) Under what conditions is it consistent?

(e) Generate $x_i \sim U(0, 10)$ for $N = 100, 500, 1000$ and, if possible, 5000. Set $\sigma^2 = 1000$. For each N, compute: (i) the conditional variances of OLS b_2 and $\widehat{\beta}_{2,mean}$; (ii) estimates of $E[(s_x^2)^{-1}]$ and $E[4/(\bar{x}_2 - \bar{x}_1)^2]$. Here $s_x^2 = N^{-1}\sum_i (x_i - \bar{x})^2$ uses divisor N.

(f) Compare the magnitudes in (e) and their changes with N. Does the alternative estimator appear consistent?

(g) Explain the large-sample limits 0.12 for the inverse-variance quantity and 0.16 for the squared-gap quantity in (e)(ii). Hint: $\text{var}[U(a, b)] = (b - a)^2/12$.

(h) Would the estimator be consistent if observations were assigned randomly to the two halves instead of sorted by x? Explain.

Exercise 5.21: Rainfall and technological change

Using `toody5`, estimate $Y_t = \beta_1 + \beta_2 TREND_t + \beta_3 RAIN_t + \beta_4 RAIN_t^2 + \beta_5 RAIN_t TREND_t + e_t$. Yield Y is tonnes per hectare in Toodyay, Western Australia. $TREND = 0, 0.1, \ldots, 4.7$ for 1950-1997. Growing-season rainfall is in decimeters (the book uses approximately 4 inches per decimeter).

(a) Report coefficient estimates, standard errors, t statistics, and p-values in a table.

(b) Which estimates differ from zero at (i) 5% and (ii) 10%?

(c) Do the coefficients have the expected signs? Explain, considering that technological progress may produce drought-resistant varieties.

(d) Find point estimates and 95% confidence intervals for the marginal effect of rainfall in (i) 1959, with $RAIN = 2.98$, and (ii) 1995, with $RAIN = 4.797$. Discuss.

(e) Find point estimates and 95% confidence intervals for rainfall that maximizes expected yield in (i) 1959 and (ii) 1995. Discuss.

Hints for Exercise 5.21

The student R worksheet downloads `toody5` in `START UP`. Use `head(toody5)` and `names(toody5)` to inspect the data. The values of `trend` are 0.9 in 1959 and 4.5 in 1995.

(a) Estimate the model with a squared rainfall term and an interaction:

```
model_521 <- lm(
  y ~ trend + rain + I(rain^2) + trend:rain,
  data = toody5
)

coef(summary(model_521))
```

`I(rain^2)` tells R to use the numerical square of rainfall. The coefficient table contains the estimates, standard errors, t-statistics, and p-values.

(b) The reported p-values are for two-sided tests of $H_0 : \beta_k = 0$. Compare each p-value first with 0.05 and then with 0.10. A coefficient can fail to be significant at 5% but still be significant at 10%.

(c) Consider the model as a curved rainfall response that may change over time. A positive coefficient on $RAIN$ and a negative coefficient on $RAIN^2$ allow diminishing returns and eventually a decline in yield when rainfall is very high. The rainfall–trend interaction shows whether the marginal rainfall effect changes as technology develops. Interpret combinations of coefficients, not each term in isolation.

(d) Differentiate the conditional mean with respect to rainfall:

$$\frac{\partial E(Y_t \mid \mathbf{X}_t)}{\partial RAIN_t} = \beta_3 + 2\beta_4 RAIN_t + \beta_5 TREND_t.$$

Insert $(RAIN, TREND) = (2.98, 0.9)$ for 1959 and $(4.797, 4.5)$ for 1995. The expression is a linear combination of coefficients. You can obtain its estimate and standard error with `car::deltaMethod()`:

```
car::deltaMethod(
  model_521,
  "b3 + 2*b4*2.98 + b5*0.9",
  parameterNames = c("b1", "b2", "b3", "b4", "b5")
)
```

Change the rainfall and trend values for 1995. Form each interval as

$$\widehat{m} \pm t_{0.975,\, N-K}\, se(\widehat{m}).$$

(e) Expected yield is maximized where the marginal rainfall effect from part (d) equals zero. At trend value T, solve for rainfall:

$$RAIN^*(T) = -\frac{\beta_3 + \beta_5 T}{2\beta_4}.$$

Check the sign of β_4 to determine whether the turning point is a maximum. Estimate the nonlinear expression and its delta-method standard error with, for example,

```r
car::deltaMethod(
  model_521,
  "-(b3 + b5*0.9)/(2*b4)",
  parameterNames = c("b1", "b2", "b3", "b4", "b5")
)
```

Replace 0.9 with 4.5 for 1995, then construct the approximate 95% intervals. Because the turning point is a nonlinear ratio of coefficients, these intervals use a large-sample delta-method approximation.

Exercise 5.22: Climate and technological change

Use `toody5` and the model $Y_t = \beta_1 + \beta_2 TREND_t + \beta_3 RAIN_t + \beta_4 RAIN_t^2 + \beta_5 RAIN_t TREND_t + e_t$. Yield is tonnes per hectare; rainfall is in decimeters; $TREND = 0, 0.1, \ldots, 4.7$ for 1950-1997.

(a) Report estimates, standard errors, t statistics, and p-values.

(b) For 1974 ($TREND = 2.4$, $RAIN = 4.576$), test at 5% that extra rainfall does not increase expected yield against a positive effect.

(c) Hold rainfall at its median, 3.8355. Find a point estimate and 95% confidence interval for the expected yield improvement attributable to technological change from 1960 to 1995.

(d) Estimate $RAIN = \alpha_1 + \alpha_2 TREND + e$. At 5%, test no decline in mean rainfall against declining mean rainfall.

(e) Predict mean rainfall in 1960 and 1995 using (d).

(f) Hold $TREND_{1995} = TREND_{1960}$, so there is no technological change. Using (e), estimate the decline in mean yield attributable to changing rainfall from 1960 to 1995.

(g) Hold $E(RAIN_{1995}) = E(RAIN_{1960})$, so there is no rainfall change. Estimate the increase in mean yield attributable to technology.

(h) Compare the estimates from (c), (f), and (g).

Exercise 5.23: Quantity discounts and quality premia

The `cocaine` data contain 56 sales in northeastern California, 1984-1991. $PRICE$ is dollars per gram, $QUANT$ grams sold, $QUAL$ percentage purity, and $TREND = 1, \ldots, 8$ for 1984-1991. Consider $PRICE = \beta_1 + \beta_2 QUANT + \beta_3 QUAL + \beta_4 TREND + e$.

(a) What signs do you expect for β_2, β_3, and β_4?

(b) Estimate the model, report the results and interpret the coefficients. Do the signs match your expectations?

(c) What fraction of price variation is jointly explained by quantity, quality and time?

(d) Sellers may accept a lower unit price for larger sales because fewer transactions reduce their exposure to detection. State suitable null and alternative hypotheses and test for a quantity discount.

(e) Test that quality has no effect on expected price against a positive quality premium.

(f) What is the estimated average annual change in price? Suggest an explanation for its direction.

Exercise 5.24: House size and age categories

The `collegetown` data cover 500 Baton Rouge houses sold in 2009-2013. $PRICE$ is thousands of dollars, $SQFT$ hundreds of square feet, and AGE a category from 1 (newest) to 11 (oldest). Use the age definitions in the publisher's `collegetown.def` file. Estimate $PRICE = \beta_1 + \beta_2 SQFT + \beta_3(SQFT \times AGE) + e$.

(a) Report results. Do the coefficients differ significantly from zero?

(b) Derive the marginal effects with respect to $SQFT$ and AGE. Interpret the coefficients. What assumptions follow from treating the ordered age categories this way?

(c) Find point estimates and 95% confidence intervals for the size effect at ages (i) 5, (ii) 20, and (iii) 40 years, using the correct age codes. How does the effect change with age?

(d) For houses of (i) 1500, (ii) 3000, and (iii) 4500 square feet, test at 5% that moving one age category higher reduces expected price by 6000 dollars, against a two-sided alternative. Discuss.

(e) Find a 95% prediction interval for a 60-year-old, 2500-square-foot house. Identify the four observations in its age category with floor space from 2450 to 2550 square feet. What were their prices? How many lie inside the interval? Assess the model's forecasting usefulness.

Exercise 5.25: House prices in a nonlinear log model

Use all 500 observations in `collegetown`. $PRICE$ is thousands of dollars and $SQFT$ hundreds of square feet. Estimate $\ln(PRICE) = \beta_1 + \beta_2 SQFT + \beta_3\sqrt{SQFT} + e$. Assume MR1-MR6, including $e \mid SQFT \sim N(0, \sigma^2)$.

(a) Report results and assess coefficient significance.

(b) Derive $\partial E[\ln(PRICE) \mid SQFT]/\partial SQFT$. Discuss its form and the expected signs of β_2 and β_3.

(c) Find and interpret point estimates and 95% confidence intervals for that derivative at 1500, 3000, and 4500 square feet.

(d) Show that $\partial E(PRICE \mid SQFT)/\partial SQFT = S \times C$, where $S = [\beta_2 + \beta_3/(2\sqrt{SQFT})]\exp(\beta_1 + \beta_2 SQFT + \beta_3\sqrt{SQFT})$ and $C = \exp(\sigma^2/2)$. A hat denotes replacement by estimated parameters.

(e) Estimate $S \times C$ for each size in (c).

(f) Find standard errors for $\hat{S}$ at each size, initially avoiding the additional uncertainty in $\hat{C}$.

(g) Using estimates from (e) and standard errors from (f), test at 5% whether the price derivative equals 9 against a two-sided alternative, at each size. State your conclusions.

(h), optional. Given $\text{var}(\hat{\sigma}^2) = 2\sigma^4/(N-K)$, estimate $\text{var}(\hat{C})$. The book states that $\hat{S}$ and $\hat{C}$ are independent. Use $\widehat{\text{var}}(\hat{S}\hat{C}) = \hat{S}^2\widehat{\text{var}}(\hat{C}) + \hat{C}^2\widehat{\text{var}}(\hat{S}) + \widehat{\text{var}}(\hat{C})\widehat{\text{var}}(\hat{S})$ to find standard errors for the product. Compare with (f). Would the tests in (g) change?

Exercise 5.26: Growth, inflation and presidential votes

Use all observations in `fair5`. $VOTE$ is the Democratic percentage of the two-party vote. $GROWTH$ and $INFLAT$ are economic growth and inflation multiplied by +1 under Democratic incumbency and -1 under Republican incumbency (see the definitions accompanying `fair5`). Estimate $VOTE = \beta_1 + \beta_2 GROWTH + \beta_3 INFLAT + e$.

(a) Report results. Are β_2 and β_3 different from zero at 10%? Did you use one- or two-tailed tests? Explain.

(b) With inflation 3% and Democratic incumbency, predict both parties' vote shares for economic growth of (i) -2%, (ii) 0%, and (iii) 3%.

(c) Repeat (b) under Republican incumbency.

(d) Do the predictions suggest a baseline popular-vote advantage for Democrats or Republicans?

(e) Compare two scenarios, both with economic growth and inflation of 2%: (1) Democratic incumbency and (2) Republican incumbency. At 5%, test whether the expected Democratic share in scenario 1 equals the expected Republican share in scenario 2.

Exercise 5.27: Painting age, area and auction prices

Use `ashcan_small`, keeping only sold paintings ($SOLD = 1$). $RHAMMER$ is sale price in thousands of dollars, $YEARS_OLD$ time from completion to auction, and $INCHSQ$ area in square inches. Create $INCHSQ10 = INCHSQ/10$.

(a) Estimate and report $RHAMMER = \beta_1 + \beta_2 YEARS_OLD + \beta_3 INCHSQ10 + e$.

(b) Find a point estimate and 95% confidence interval for expected yearly appreciation in price.

(c) Find a point estimate and 95% confidence interval for the extra value associated with 10 additional square inches.

(d) Add $INCHSQ10^2$ and re-estimate. Report results and explain why a squared term could be useful.

(e) How much does this change the interval in (b)?

(f) Using the quadratic model, find 95% confidence intervals for the marginal value of 10 additional square inches at areas (i) 50, (ii) 250, and (iii) 900 square inches. Discuss the pattern.

(g) Find a 95% confidence interval for the area that maximizes price.

(h) Find a 95% confidence interval for expected price of a 75-year-old, 100-square-inch painting.

(i) How long must a 100-square-inch painting be kept before its expected price becomes positive?

Exercise 5.28: Painting prices in a log model

Use `ashcan_small`, keeping $SOLD = 1$. $RHAMMER$ is price in thousands of dollars, $YEARS_OLD$ the painting's age, and $INCHSQ$ its area. Create $INCHSQ10 = INCHSQ/10$.

(a) Estimate and report $\ln(RHAMMER) = \beta_1 + \beta_2 YEARS_OLD + \beta_3 INCHSQ10 + e$.

(b) Find yearly percentage appreciation and a 95% confidence interval.

(c) At 5%, test that 10 additional square inches increase value by 2% or less against an increase greater than 2%.

(d) Add $INCHSQ10^2$, re-estimate, report results and explain why the additional term may be appropriate.

(e) How much does this change the interval in (b)?

(f) Repeat (c) for areas (i) 50, (ii) 250, and (iii) 900 square inches. What do you observe?

(g) Find a 95% confidence interval for the area that maximizes price.

(h) Find a 95% interval for the price quantity requested in the book for a 75-year-old, 100-square-inch painting. Use $\exp\{E[\ln(RHAMMER) \mid \widehat{YEARS_OLD} = 75, INCHSQ10 = 10]\}$ and its standard error. Explain what price quantity this estimator measures.

Exercise 5.29: Crime and punishment

Use `crime`, a panel of 90 North Carolina counties for 1981-1987, and select **1986 only**. Estimate a model relating $LCRMRTE$, the log crime rate, to:

- $PRBARR$: arrests divided by offenses.
- $PRBCONV$: convictions divided by arrests.
- $PRBPRIS$: prison sentences divided by convictions.
- $POLPC$: police per capita.
- $WCON$: weekly construction wages.

Write a short report addressing:

- Your expected effect of each regressor on crime.
- The fitted coefficients, their signs and statistical significance.
- Which variables appear most important for deterrence.
- How you might explain the sign of the police coefficient.

Exercise 5.30: Could Andy's advertising level be optimal?

For Big Andy's Burger Barn, use `andy` and the quadratic model $SALES = \beta_1 + \beta_2 PRICE + \beta_3 ADVERT + \beta_4 ADVERT^2 + e$. Sales and advertising expenditure are in thousands of dollars.

The book's advertising optimum satisfies $\beta_3 + 2\beta_4 ADVERT_0 = 1$.

At 5%, test whether each candidate could be optimal:

(a) $ADVERT_0 = 1.75$.

(b) $ADVERT_0 = 1.90$.

(c) $ADVERT_0 = 2.30$.

Report a p-value for each test.

Exercise 5.31: Bill's journey to work

The `commute5` data describe 249 working days in 2015. Bill leaves Carnegie between 6:30 and 8:00 AM to drive to the University of Melbourne. $TIME$ is travel time in minutes, $DEPART$ minutes after 6:30 AM, $REDS$ red lights encountered, and $TRAINS$ trains waited for.

(a) Estimate $TIME = \beta_1 + \beta_2 DEPART + \beta_3 REDS + \beta_4 TRAINS + e$. Report and interpret every coefficient, including the intercept.

(b) Find 95% confidence intervals for all coefficients. Are the estimates precise?

(c) At 5%, test that expected delay per red light is at least 2 minutes against less than 2 minutes.

(d) At 10%, test that expected delay per train equals 3 minutes against a two-sided alternative.

(e) Holding other variables fixed, test at 5% that leaving at 7:30 rather than 7:00 AM adds at least 10 minutes to expected travel time against an increase of less than 10 minutes.

(f) At 5%, test that expected delay per train is at least three times the delay per red light against a smaller multiple.

(g) With six red lights and one train, test at 5% that a 7:00 AM departure gets Bill to the university by 7:45 AM in expected-time terms, against the alternative that it does not.

(h) If being late is unacceptable, are the hypotheses in (g) the appropriate way round? What happens if they are reversed?

Exercise 5.32: Partial out lights and trains

Use `commute5` and the model from Exercise 5.31: $TIME = \beta_1 + \beta_2 DEPART + \beta_3 REDS + \beta_4 TRAINS + e$. Bill is mainly interested in β_2: he controls departure time, but not the red lights or trains.

(a) Regress $DEPART$ on $REDS$ and $TRAINS$, including an intercept, and save the residuals. Which coefficients differ from zero at 5%? Do any significant relationships have a causal interpretation?

(b) Regress $TIME$ on $REDS$ and $TRAINS$, including an intercept, and save the residuals. Compare the light and train coefficients with those in the full model that includes departure time.

(c) Use the two sets of residuals to estimate β_2. Adjust the regression output to obtain its correct standard error.

Exercise 5.33: Education, experience and log wages

Using `cps5_small`, estimate $\ln(WAGE) = \beta_1 + \beta_2 EDUC + \beta_3 EDUC^2 + \beta_4 EXPER + \beta_5 EXPER^2 + \beta_6(EDUC \times EXPER) + e$. Education and experience are in years.

(a) At what significance levels do the coefficients differ from zero?

(b) Derive the marginal effect of education on expected log wage. How does its estimate change with education and experience?

(c) Calculate that effect for every sample observation and draw a histogram. Discuss; report its median, 5th and 95th percentiles.

(d) Derive the marginal effect of experience on expected log wage. How does its estimate change with education and experience?

(e) Repeat (c) for the experience effect.

(f) David has 17 years of education and 8 years of experience; Svetlana has 16 and 18. At 5%, test that Svetlana's expected log wage is at least David's against the alternative that David's is higher. State the hypotheses in terms of the parameters.

(g) Repeat the comparison after both gain eight years of experience and no further education. Is the conclusion unchanged? Explain.

(h) Wendy has 12 years of education and 17 of experience; Jill has 16 and 11. At 5%, test equality of their marginal experience effects against a two-sided alternative. State the parameter restrictions.

(i) How many additional years of experience must Jill gain before her marginal experience effect becomes negative? Find a 95% confidence interval for this quantity.